Runtime-Validated Hybrid Unitary Synthesis with Native Entangling Evolution

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We revisit the previously defined hybrid qubit-cavity target unitary $U_{\text{target}} = (I_q \otimes H_c)\,\text{CNOT}_{c\to q}\,\text{CNOT}_{q\to c}$ on the logical subspace $\{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}$ and convert an earlier symbolic-native-entangler conclusion into a runtime-backed one. Symbolically, the exact two-wait architecture remains an upper bound with process fidelity 0.9999998, while the best experimentally grounded symbolic candidate, the fully archive-driven two-wait family, reaches process fidelity 0.98568. To close the replay gap inside `cqed_sim`, we replace the exact qubit Hadamard by a replayable two-rotation decomposition, replay `FreeEvolveCondPhase` through explicit idle evolution, and replace the non-replayable local blocks by GRAPE-derived replayable surrogates. The best closed-system runtime candidate is then the surrogate-backed exact two-wait family, with process fidelity 0.90269, average probe fidelity 0.83495, average leakage 0.04318, and final Wigner overlap 0.96384 at $n_{\text{cav}} = 12$, $n_{\text{tr}} = 3$. The surrogate-backed archive-local family is close in process fidelity (0.90154) and has the higher final Wigner overlap (0.98142), but it is worse in average probe fidelity (0.81219) and nominal-noise average probe fidelity (0.58042 versus 0.61448). Runtime ranking is stable across $n_{\text{cav}} = 10, 12, 14$, but both leading runtime candidates require 4.432 µs total sequence time, so the result is an architecture-level validation rather than a hardware-readiness claim.

I. INTRODUCTION

Hybrid qubit-cavity control in circuit QED combines cheap local bosonic control with conditional and selective operations that are more expensive to calibrate and more fragile under replay [1, 2]. In a previous internal study, the target unitary

$$U_{\text{target}} = (I_q \otimes H_c)\,\text{CNOT}_{c\to q}\,\text{CNOT}_{q\to c} \tag{1}$$

was fixed on the logical basis $\{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}$, and the best native-heavy architectures were identified symbolically. That symbolic milestone established two important facts: one native dispersive wait is insufficient, while two waits are enough. What it did not establish was which of those architectures survives once every non-replayable local block is forced onto a replayable implementation.

The present follow-up answers that narrower question. We keep the same target, the same device-point conventions, and the same native-entangler-centric cost model, but we add an explicit runtime layer inside `cqed_sim` [3]. This is the critical shift: symbolic upper bounds are retained, but the final recommendation is made only after replay-backed metrics, stronger truncation checks, and saved pulse artifacts are available. The remainder of the paper describes the dispersive model and replay construction, presents the runtime shortlist and convergence results, and closes with a clear statement of what is now validated and what remains open.

II. SYSTEM AND METHODS

A. Hamiltonian

We work in the dispersive regime of a transmon-cavity system with the same effective model used in the earlier hybrid-control study. The drift Hamiltonian is

$$\frac{H_{\text{eff}}}{\hbar} = \omega_c a^\dagger a + \frac{\omega_q}{2}\sigma_z + \chi a^\dagger a\,\sigma_z + \chi' a^{\dagger 2} a^2\,\sigma_z + \frac{K}{2} a^{\dagger 2} a^2, \tag{2}$$

where a is the cavity lowering operator, σ_z is the transmon Pauli-Z operator in the logical qubit manifold, χ is the dispersive shift, χ' is the second-order dispersive shift, and K is the cavity self-Kerr. The native entangling resource of interest is free evolution under the conditional phase generated by this dispersive interaction. Local control is provided by qubit rotations, cavity displacements, and cavity-selective phases.

B. Simulation Parameters

TABLE I. System and runtime-validation parameters.

Parameter	Value	Unit
$\omega_q/2\pi$	6.150	GHz
$\omega_c/2\pi$	5.241	GHz
$\alpha/2\pi$	-255.000	MHz
$\chi/2\pi$	-2.840	MHz
$\chi'/2\pi$	-0.021	MHz
$K/2\pi$	-0.028	MHz
Native entangling time per wait	176.000	ns
Runtime reference n_{cav}	12	levels
Runtime sweep n_{cav}	10/12/14	levels
Runtime n_{tr}	3	levels
Replay grid Δt	2.000	ns
Surrogate slices	40	steps
Surrogate slice duration	32.000	ns
Qubit T_1	30.000	µs
Qubit T_2	20.000	µs
Cavity T_1	250.000	µs

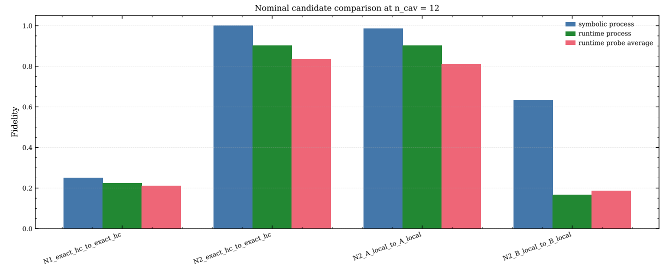

FIG. 1. Phase 5 runtime comparison. The symbolic archive-local focal candidate does not remain the runtime leader once all shortlisted candidates are forced onto replayable implementations. The exact two-wait surrogate family is the best closed-system runtime candidate, while the archive-local surrogate remains competitive and has the strongest final Wigner overlap.

C. Computational Approach

All sequence construction, optimal control, replay, and diagnostics were carried out inside `cqed_sim` [3]. The study proceeded in five logical stages. First, legacy native-heavy candidates were normalized onto a common entangling-cost bookkeeping layer. Second, a native-block search composed full-U_{target} candidates using one or two free-evolution waits. Third, symbolic depth diagnostics were generated with Bloch and Wigner checkpoints. Fourth, replay support was audited to determine which gates were directly bridge-compatible. Fifth, the symbolic shortlist was converted into a runtime shortlist.

The runtime shortlist contains four candidates: a one-wait replay baseline, a two-wait exact-surrogate family, a two-wait archive-local-surrogate family, and a direct replayable archive reference. The replay conversion used three key steps:

1. replace the exact qubit Hadamard by $R_{\hat{y}}(\pi/2)R_{\hat{x}}(\pi)$, which matches H up to global phase,
2. replay `FreeEvolveCondPhase` through an explicit zero-amplitude idle pulse segment so that the drift Hamiltonian carries the entangling action,
3. replace the non-replayable `PrimitiveGate` and `SNAP`-based local blocks by GRAPE-derived replayable surrogates.

The local surrogates were optimized at reference truncation $n_{\mathrm{cav}} = 12$, $n_{\mathrm{tr}} = 3$ using four rotating-frame control channels (storage-I, storage-Q, qubit-I, qubit-Q), 40 piecewise-constant slices, slice duration 32 ns, three random restarts, and a maximum of 60 iterations per restart. The best nominal surrogate fidelities were 0.94288 for the exact cavity-H target and 0.94762 for the archive-local target. All restarts hit the iteration limit, so these surrogates are best viewed as usable runtime stand-ins rather than converged local optima.

For each symbolic and runtime candidate we report logical process fidelity, average probe-state fidelity over six logical probes, average leakage, maximum transient cavity population, final Wigner overlap with the target probe state, and nominal-noise replay at the reference truncation. A weighted scalar cost was still computed for bookkeeping, but raw metrics were treated as primary because the sensitivity sweep shows that the scalar score can mis-rank low-fidelity candidates if the infidelity penalty is softened.

III. RESULTS

A. The Symbolic Focal Candidate Does Not Survive Replay Unchanged

Symbolically, the earlier study identified three regimes: a one-wait lower bound, an exact two-wait upper bound, and a fully archive-driven two-wait family that was the best experimentally grounded symbolic candidate. Runtime replay changes that ranking. Table II shows the reference-truncation comparison. The best replay-backed closed-system candidate is the exact two-wait surrogate family, not the archive-local one.

The leading two replay-backed candidates are very close in process fidelity, but the ordering is consistent. The exact-surrogate family reaches process fidelity 0.90269 and average probe fidelity 0.83495, while the archive-local surrogate reaches 0.90154 and 0.81219. The archive-local surrogate does, however, preserve the final Wigner structure better on the $|g, 0\rangle$ probe, with final Wigner overlap 0.98142 versus 0.96384 for the exact surrogate. This is a genuine trade-off, not numerical noise.

B. One-Wait and Direct-Replay Archive References Remain Poor

The one-wait replay baseline confirms the architectural lower bound. Even after all local pieces are replaced by replayable implementations, `R1_exact_runtime_to_exact_runtime` only reaches process fidelity 0.2234 and average probe fidelity 0.2120. It is shorter than the two-wait families, but it still fails the target badly.

The direct replayable archive reference is more revealing. The `B_local`-based family requires no surrogate for its local block, yet it performs substantially worse: process fidelity 0.16665, average probe fidelity 0.18729, leakage 0.14324, and runtime duration 7.912 µs. This confirms that bridge compatibility alone is not enough to justify a candidate. The replayable archive path remains a useful reference, but not a competitive synthesis route.

TABLE II. Reference-truncation comparison at $n_{\text{cav}} = 12$, $n_{\text{tr}} = 3$. Symbolic metrics are taken from the corresponding symbolic reference candidate. Runtime metrics are from replay-backed evaluation.

Runtime candidate	Symbolic reference	F_{sym}	F_{run}	$\overline{F}_{\text{probe,run}}$	L_{run}	$\overline{F}_{\text{probe,noise}}$	t_{run} (µs)
R1 exact-runtime	N1 exact	0.250 000	0.223 401	0.211 970	0.028 603	0.048 006	2.856
R2 exact-runtime	N2 exact	1.000 000	0.902 693	0.834 953	0.043 177	0.614 483	4.432
R2 A-runtime	N2 A-local	0.985 681	0.901 541	0.812 195	0.047 255	0.580 417	4.432
R2 B-replay	N2 B-local	0.634 498	0.166 650	0.187 290	0.143 236	0.048 845	7.912

C. Noise Exposure Now Dominates the Practical Interpretation

The strongest replay-backed candidates are not limited by native entangling time; they are limited by local-surrogate duration. Each sequence contains three 1.28 µs surrogate blocks plus two native waits and two short qubit-H replacements, for a total runtime of 4.432 µs. Under the nominal noise model inherited from the earlier hybrid study, the exact surrogate falls from average probe fidelity 0.83495 in the closed system to 0.61448, while the archive-local surrogate falls to 0.58042. The runtime study therefore validates the architecture and the ranking, but it does not yet validate an experimentally ready pulse schedule.

IV. VALIDATION

A. Sanity Checks

Three sanity checks pass. First, the symbolic one-wait family remains pinned near 0.25, confirming that one native wait is structurally insufficient. Second, the symbolic exact two-wait family still reaches unit fidelity to numerical precision, confirming the architecture-level identity on the logical subspace. Third, the replay-backed one-wait baseline also remains poor, so the runtime follow-up does not reopen the one-wait question.

B. Convergence Analysis

The user explicitly requested stronger truncation checks, so the runtime shortlist was replayed at $n_{\text{cav}} = 10$, 12, and 14 with $n_{\text{tr}} = 3$. Table III reports the spans of the main runtime metrics over that sweep. All spans are small enough that the ranking shift from the symbolic archive-local candidate to the replay-backed exact surrogate cannot be attributed to truncation error.

The convergence figure in Fig. 2 shows the same result graphically: the runtime metrics are effectively flat across the requested truncation sweep.

TABLE III. Metric spans over the runtime truncation sweep $n_{\text{cav}} = 10, 12, 14$.

Candidate	ΔF_{proc}	$\Delta\overline{F}_{\text{probe}}$	ΔL	ΔO_W
R1 exact-runtime	2.53×10^{-6}	1.30×10^{-6}	6.80×10^{-6}	3.44×10^{-6}
R2 exact-runtime	7.67×10^{-6}	9.80×10^{-6}	1.20×10^{-5}	7.87×10^{-6}
R2 A-runtime	2.79×10^{-5}	6.84×10^{-5}	6.89×10^{-5}	1.69×10^{-4}
R2 B-replay	1.30×10^{-6}	2.25×10^{-6}	6.46×10^{-6}	3.30×10^{-6}

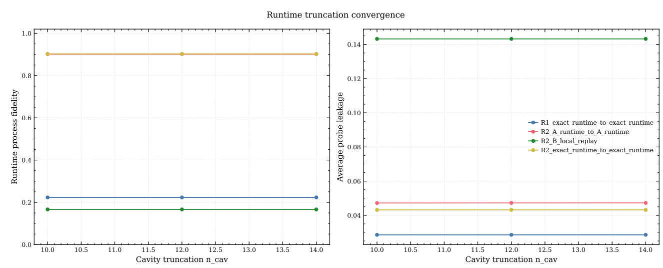

FIG. 2. Runtime convergence across $n_{\text{cav}} = 10, 12, 14$. Metric spans are small for every replay-backed candidate, so the ranking shift is not a truncation artifact.

C. Literature Comparison

There is no external published benchmark for this internal target unitary, so the relevant comparison is internal rather than literature-facing: symbolic upper bound versus replay-backed realizations under a common device point and common bookkeeping layer.

V. DISCUSSION

The main scientific outcome is not that symbolic analysis was wrong; it is that symbolic and runtime notions of "best candidate" are meaningfully different. Symbolically, the fully archive-driven two-wait family was the most compelling experimentally grounded candidate because it preserved the correct architecture without relying on ideal cavity-H references. Once every candidate is forced onto a replayable implementation, that ordering changes. The best replay-backed closed-system candidate is the exact two-wait family with replayable surrogates for the non-replayable local blocks.

At the same time, the replay-backed archive-local surrogate is not simply worse. It retains the strongest final

Wigner overlap on the probe that is most sensitive to cavity-structure preservation. This suggests that the two leading runtime families are emphasizing different physical virtues: the exact surrogate wins on logical action, while the archive-local surrogate preserves one slice of bosonic structure slightly better. That difference is small enough to justify keeping both candidates alive for future optimization, but the present recommendation still goes to the exact-surrogate family because it wins on the primary logical metrics and on nominal-noise average probe fidelity.

The more serious practical issue is duration. The native entangling resource is cheap compared with the replayable local surrogates, so the present runtime winner is not limited by entangling exposure. It is limited by local-control overhead. This is why the nominal-noise numbers are mediocre despite reasonably strong closed-system fidelities. The correct conclusion is therefore skeptical but useful: the two-wait native architecture is validated, the runtime ranking is resolved, and the bottleneck has shifted to local replayable control quality and duration.

The weighted scalar cost deserves one final caution. In the sensitivity sweep, reducing the infidelity weight by 25 percent makes the one-wait replay baseline look cheapest even though it plainly fails the target. That is not a scientific surprise; it is a bookkeeping warning. Any future decision rule should report a Pareto frontier or enforce a minimum acceptable fidelity before cost is used to break ties.

VI. CONCLUSION

This follow-up converts the earlier symbolic conclusion into a runtime-backed one and yields four concrete conclusions.

First, one native free-evolution wait remains an architectural lower bound both symbolically and under replay. Second, the exact two-wait architecture remains the symbolic upper bound. Third, the best experimentally grounded symbolic candidate does not remain the runtime leader once replay is enforced: the best replay-backed closed-system candidate is the surrogate-backed exact two-wait family, not the surrogate-backed archive-local family. Fourth, the present runtime winner is not hardware-ready because its 4.432 µs duration drives nominal-noise average probe fidelity down to 0.61448.

The final recommendation is therefore precise. The correct architecture for this target is the two-native-wait family, and the current replay-backed focal candidate is `R2_exact_runtime_to_exact_runtime`. The next study should optimize local replayable controls, not reopen the symbolic architecture search.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

The runtime-validation gap is closed, but the local replayable surrogates are not converged. All GRAPE restarts stopped at the iteration limit, and the best local surrogate fidelities are only 0.94288 and 0.94762. This means the present runtime winner is best within the current surrogate family, not necessarily best within all replayable local-control strategies.

A second limitation is that noise was only evaluated at one nominal operating point and one truncation. The nominal-noise calculation is enough to show that duration is now the dominant practical problem, but it is not enough to claim robustness under parameter drifts, thermal population, or stronger open-system effects.

A third limitation is methodological. The scalar weighted cost is convenient for bookkeeping, but it can mis-rank bad candidates unless a fidelity floor is imposed. The raw process, probe, leakage, and Wigner metrics should therefore remain the decision metrics in any future extension.

B. Suggested Improvements

- **[P1 — HIGH]** Re-optimize the replayable local surrogates with longer optimizer budgets, shorter-duration ansatzes, or smoother parameterizations. The present runtime winner is limited by three 1.28 µs surrogate blocks per sequence.
- **[P1 — MEDIUM]** Add a fidelity floor or Pareto filter before scalar cost ranking. The current sensitivity sweep shows that cost alone can prefer the one-wait baseline under weak fidelity weighting.
- **[P2 — MEDIUM]** Expand the noise and robustness study to include amplitude, duration, and dispersive-parameter perturbations in addition to the nominal T_1/T_2 point used here.
- **[P2 — MEDIUM]** Compare the current GRAPE surrogates against a direct model-backed `SNAP` runtime path if one can be added inside `cqed_sim` without violating the framework-first rule.

C. Open Questions

Two open questions stand out. Why does the archive-local runtime surrogate retain the higher final Wigner overlap while losing on process and average probe fidelity? And can either of the two leading runtime families be compressed below the current multi-microsecond duration without sacrificing the ~ 0.90 closed-system process-fidelity level?

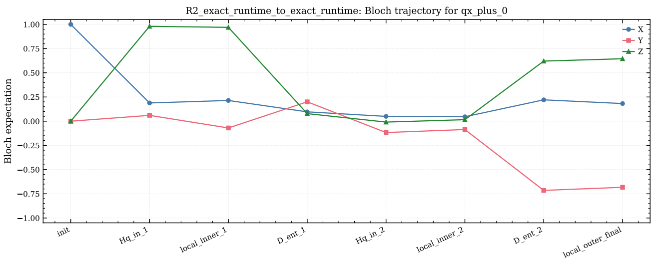

FIG. 3. Checkpointed Bloch trajectory for the replay-backed exact two-wait winner on the $|+_x, 0\rangle$ probe.

[1] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, Physical Review A **76**, 042319 (2007).

[2] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, Reviews of Modern Physics **93**, 025005 (2021).

[3] cQED Autonomous Research Platform, cqed_sim: Circuit qed simulation framework (2026).

Appendix A: Detailed Results and Data

Table IV collects the raw symbolic and runtime metrics at the reference truncation. These are the values that should be compared directly; the scalar weighted cost is only a secondary summary.

Figure 3 shows the checkpointed Bloch trajectory for the runtime winner, and Fig. 4 compares the final target and achieved Wigner functions for both leading runtime candidates.

Figure 5 records the weight-sensitivity warning. The one-wait replay baseline can become the scalar-cost winner if the infidelity term is weakened, so raw metrics must remain primary.

The machine-readable payloads are stored in `../data/phase5_runtime_validation.json`, `../data/phase5_candidate_comparison.csv`, `../data/phase5_runtime_metrics.csv`, and `../data/phase5_convergence_table.csv`.

Appendix B: Reproducibility

1. Optimized Parameters

TABLE V. Replayable local-surrogate parameters at the reference truncation.

Surrogate	F_{nom}	t (µs)	Pulses	Restarts
exact_hc runtime	0.942 88	1.28	160	3
A_local runtime	0.947 62	1.28	160	3

The replayable qubit-H replacement is fixed as $R_{\hat{y}}(\pi/2)$ followed by $R_{\hat{x}}(\pi)$, with each pulse lasting 20 ns. Each native entangler is the sequence qubit rotation – free evolution – qubit rotation with total duration 176 ns of free evolution plus two 40 ns qubit rotations.

2. Waveform and Pulse Information

The local-surrogate waveforms are stored in the machine-readable artifacts `phase5_local_surrogate_exact_hc.json`, `phase5_local_surrogate_exact_hc_grape.json`, `phase5_local_surrogate_A_local.json`, and `phase5_local_surrogate_A_local_grape.json` in the `../artifacts/` directory. Each surrogate uses four exported control channels, 40 time slices, and square-envelope pulses on the replay grid.

3. Gate Sequence and Decomposition

The runtime winner `R2_exact_runtime_to_exact_runtime` uses the ordered segment sequence

1. replayable qubit-H,
2. exact-H local surrogate,
3. native entangler,
4. replayable qubit-H,
5. exact-H local surrogate,
6. native entangler,
7. exact-H local surrogate.

The archive-local runtime candidate uses the same layout but substitutes the archive-local surrogate for the three local segments. Exact serialized segment metadata are stored in `phase5_runtime_candidate_R2_exact_runtime_to_exact_runtime.` and `phase5_runtime_candidate_R2_A_runtime_to_A_runtime.json`

4. Modeling and Simulation Assumptions

Symbolic candidates were evaluated on the logical subspace of the dispersive model using the same device point as the earlier hybrid study. Runtime replay used $n_{\mathrm{tr}} = 3$ and $n_{\mathrm{cav}} = 10$, 12, and 14, a replay time step of 2 ns, and nominal noise only at the reference truncation. The noise model uses qubit $T_1 = 30\,\mu\mathrm{s}$, qubit $T_2 = 20\,\mu\mathrm{s}$, pure dephasing inferred from those values, and cavity decay $\kappa = 1/T_{1,c}$ with $T_{1,c} = 250\,\mu\mathrm{s}$.

TABLE IV. Raw candidate metrics at $n_{\text{cav}} = 12$, $n_{\text{tr}} = 3$.

Candidate	F_{sym}	F_{run}	$\overline{F}_{\text{probe,run}}$	L_{run}	$\overline{F}_{\text{probe,noise}}$	O_W	t_{run} (µs)
R1 exact-runtime	0.250 000	0.223 401	0.211 970	0.028 603	0.048 006	0.512 031	2.856
R2 exact-runtime	1.000 000	0.902 693	0.834 953	0.043 177	0.614 483	0.963 835	4.432
R2 A-runtime	0.985 681	0.901 541	0.812 195	0.047 255	0.580 417	0.981 417	4.432
R2 B-replay	0.634 498	0.166 650	0.187 290	0.143 236	0.048 845	0.259 742	7.912

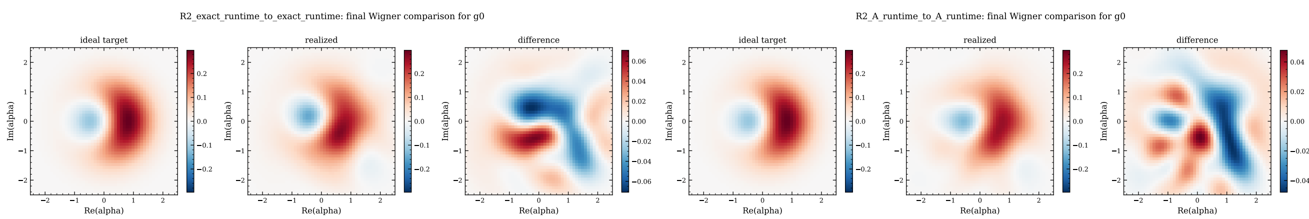

FIG. 4. Final Wigner comparison on the $|g, 0\rangle$ probe for the two leading replay-backed candidates. The archive-local surrogate retains slightly better final Wigner overlap, even though the exact-surrogate family wins on the main logical fidelities.

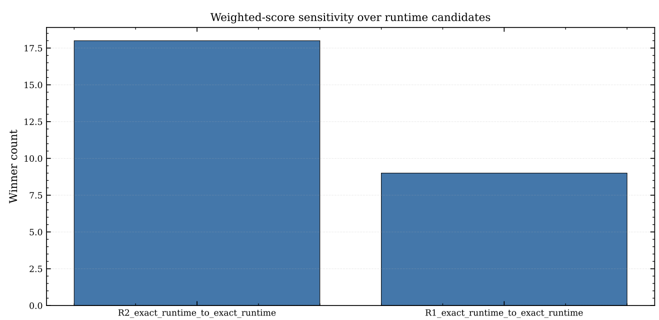

FIG. 5. Weight-sensitivity sweep for the scalar cost. The one-wait baseline can become the scalar-cost winner if the fidelity term is weakened, which is why the study reports raw metrics rather than relying on a single weighted score.

5. Reproduction Procedure

To reproduce the Phase 5 results from the study folder:

1. run `python scripts/phase5_runtime_validation.py` from `studies/hybrid_unitary_native_entangling_evolution/`;
2. confirm that the script writes the Phase 5 CSV tables into `data/`, the surrogate and runtime-decomposition artifacts into `artifacts/`, and the runtime comparison, convergence, Bloch, and Wigner figures into `figures/`;
3. verify agreement by checking that the leading runtime candidate is `R2_exact_runtime_to_exact_runtime` with process fidelity near 0.90269 and that the convergence spans in `phase5_convergence_table.csv` remain at or below the values reported in Table III.

6. Saved Artifacts

The key machine-readable artifacts are:

- `phase5_local_surrogate_exact_hc.json`: exported replayable pulses and metadata for the exact-H local surrogate,
- `phase5_local_surrogate_exact_hc_grape.json`: full GRAPE payload for that surrogate,
- `phase5_local_surrogate_A_local.json`: exported replayable pulses and metadata for the archive-local surrogate,
- `phase5_local_surrogate_A_local_grape.json`: full GRAPE payload for that surrogate,
- `phase5_runtime_candidate_R2_exact_runtime_to_exact_run` serialized replay decomposition for the runtime winner,
- `phase5_runtime_candidate_R2_A_runtime_to_A_runtime.jso` serialized replay decomposition for the archive-local runtime finalist,
- `phase5_runtime_candidate_R2_B_local_replay.json`: serialized direct replay decomposition for the archive reference.

These files are sufficient for a future agent to inspect the runtime candidate structure, reload pulse metadata, and reproduce the ranking without reverse engineering the report text.